DEVVRATH TEWATIA

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EDUCATION

Indian Institute of Technology Patna — B.Tech, Mechanical Engineering *2023–2027*
CPI: 7.9/10 | Class XII (CBSE): 95%, Modern Delhi International School

TECHNICAL SKILLS

Programming: Python, C++, SQL
Robotics: ROS 2, Gazebo, kinematics and dynamics, inverse kinematics, collision checking, motion planning
ML & Computer Vision: PyTorch, TensorFlow/Keras, OpenCV, CNNs, supervised learning, SHAP, anomaly detection
Software & Deployment: Git/GitHub, automated testing, continuous integration, FastAPI, Streamlit, AWS IoT Core
Core Engineering: SolidWorks, ANSYS Workbench

SELECTED PROJECTS

7-DOF Bimanual Mobile Manipulation (ROS 2, Gazebo) *B.Tech Project | GitHub*

- Built a simulated dual-arm mobile manipulator (7-DOF per arm, Kobuki base) integrating perception, localisation, navigation, collision-aware planning, bimanual grasping, reorientation and placement.
- Developed a closed-form 7-DOF inverse-kinematics solver with elbow-swivel selection; achieved collision-free 90° reorientation with 13–14 mm clearance, up to 5.4° tracking error and 1–4 mm object drift in demonstrated runs.
- Built a 166-check validation harness; mutation testing caught 12 of 13 injected faults.

Robot Dynamics Solver *Independent Project | GitHub*

- Implemented a generic N-DOF rigid-body dynamics library with Lagrangian and Featherstone backends (RNEA, CRBA, ABA), quintic trajectories, RK4 simulation, and PD, feedforward and computed-torque control.
- Cross-validated both formulations to within 1.6×10^{-14} across 13 models (2–9 DOF), using analytical and SymPy Euler–Lagrange checks; added 316 tests and CI.
- Computed-torque control reduced 2-link tracking error from 5.4×10^{-2} rad to 1.6×10^{-11} rad; RK4 energy drift was $\sim 4 \times 10^{-9}$ versus 7×10^{-2} for explicit Euler.

Metal Surface Defect Classification using CNN *Computer Vision · TensorFlow/Keras*

- Developed a 305K-parameter CNN for 1,800 grayscale metal-surface images across six defect classes, achieving 99.06% ± 0.87% mean accuracy in stratified, duplicate-grouped 5-fold cross-validation; assessed macro-F1 and class-wise errors.
- Used batch normalisation, dropout, early stopping and adaptive learning-rate scheduling; confusion-matrix analysis identified errors between Patches/Crazing and Pitted/Inclusion classes.
- Built a reusable single-image inference tool returning predicted defect classes and top-k softmax scores.

Energy Optimisation in Metal Manufacturing *Simulated Model | GitHub*

- Built a CNC milling simulator and physics-based power model ($R^2 = 0.93$, MAPE = 3.3% on unseen simulated jobs); used SHAP to interpret energy drivers and residuals for anomaly detection.
- Designed a cycle-time-constrained optimiser with 37% mean simulated energy savings; served predictions and simulated telemetry through FastAPI, AWS IoT Core (MQTT) and Streamlit.

Autonomous UAV Landing on a Moving Train *Coursework Technical Review*

- Co-authored a literature review on UAV take-off and landing on moving trains, covering GNSS-denied navigation, LiDAR–visual–inertial odometry, MPC, ADRC and safety constraints; review only, no implementation.

POSITIONS OF RESPONSIBILITY

- Chassis Design Coordinator, IIT Patna Motorsports: Built the structural model in SolidWorks and finalised seat mounting within packaging constraints.
- Coordinator, Student Council of Mechanical Engineering: Helped run EFX: Upkarna workshops and CAD events, and coordinated the Robo Wars competition with Celesta.

ACHIEVEMENTS & RESEARCH INTERESTS

Achievements: Amazon ML Challenge 2025 — Top 500 | JEE Main — AIR 8,221 | Top 15% of batch by CPI
Research interests: Physics-informed ML, learning for manipulation and control, imitation and reinforcement learning, motion planning, sim-to-real transfer.